Mid-tern, Catch-up

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https://github.com/hyunneee/hyunneee.github.io

in [VMP_midterm catch-up] branch

2. A clock having two needles (hour and minute)

- Can you make the clock synchrozed with your computer clock?

```python
from datetime import datetime 현재시간을 불러오기 위한 명령.
import numpy as np
import cv2

height, width = 1000, 1500
# now = datetime.now()
# hour = now.hour
# min = now.minute

def getline(x0,y0,x1,y1):
    points = []
    if abs(x1-x0)>=abs(y1-y0):
        n = abs(x1-x0)
    else:
        n = abs(y1-y0)
    dx = (x1-x0)/n
    dy = (y1-y0)/n
    x,y = x0,y0
    for k in range(n):
        points.append((int(x),int(y),1))
        x = x+dx
        y = y+dy
    return points

def drawLine(canvas, x0, y0, x1, y1, color=(255, 255, 255)):
    xys = getline(x0, y0, x1, y1)
    for xy in xys:
        x, y, a= xy
        canvas[y, x, :] = color
    return

def drawPolygon(canvas, pts, color, axis=False):
    n = pts.shape[0]
    for k in range(n):
        drawLine(canvas, pts[int(k%n),0], pts[int(k%n),1],
                 pts[int((k+1)%n),0], pts[int((k+1)%n),1], color)

    if axis == True: # center - pts[0]
        center = np.array([0., 0, 0])
        for p in pts:
            center += p
        center = center / pts.shape[0]
        center = center.astype('int')
        # print(center)
        drawLine(canvas, center[0],center[1], pts[0][0],pts[0][1], color=(255, 128, 128))
    #
    return

def deg2rad(deg):
    rad = deg * np.pi / 180.
    return rad

def drawPolygon(canvas, pts, color, axis=False):
    n = pts.shape[0]
    # print(type(shape))
    for k in range(n):
        drawLine(canvas, pts[int(k%n),0], pts[int(k%n),1],
                 pts[int((k+1)%n),0], pts[int((k+1)%n),1], color)

    if axis == True: # center - pts[0]
        center = np.array([0., 0,0])
        for p in pts:
            center += p
        center = center / pts.shape[0]
        center = center.astype('int')
        # print(center)
        drawLine(canvas, center[0],center[1], pts[0][0],pts[0][1], color=(255, 128, 128))
    #
    return
```

① 다 동일

①

```python
def getRegularNGon(ngon):
    delta = 360. / ngon
    points = []
    for i in range(ngon):
        degree = i * delta
        radian = deg2rad(degree)
        x =   np.cos(radian)
        y =   np.sin(radian)
        points.append( (x, y,1) )
    #
    points = np.array(points)
    return points

def makeTmat(a,b):
    T = np.eye(3,3)
    T[0,2] = a
    T[1,2] = b
    # print(T)
    return T

def makeRmat(deg):
    rad = deg2rad(deg)
    c = np.cos(rad)
    s = np.sin(rad)
    R =  np.eye(3,3)
    R[0,0] = c
    R[0,1] = -s
    R[1,0] = s
    R[1,1] = c
    # print(R)
    return R

def getrectangle(len):
    points = []
    points.append((0,0,1))
    points.append((3*len,0,1))
    points.append((3*len,len,1))
    points.append((0,len,1))

    points = np.array(points)
    # print(points)
    return points
```

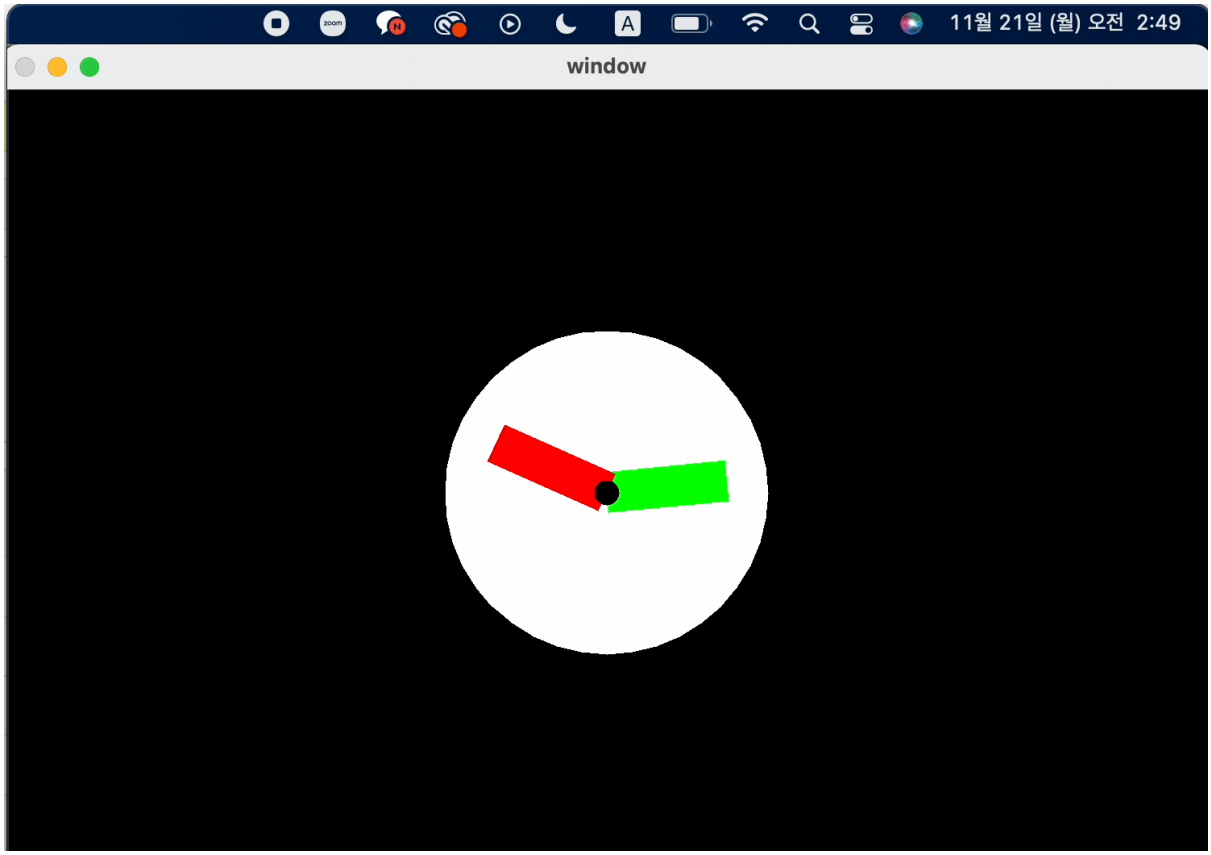
길이를 변수로 갖고
가로가 len×3 이 세로가 len인
사각형 만들기

```python
def drawColored(canvas,points,color):
    n = points.shape[0]
    pts = []

    for k in range(n):
        a = getline(points[int(k%n),0], points[int(k%n),1],points[int((k+1)%n),0], points[int((k+1)%n),1])
        pts.extend(a)
    pts = np.array(pts)
    xs = pts[:,0]
    ys = pts[:,1]
    xmin = min(xs)
    xmax = max(xs)
    ymin = min(ys)
    ymax = max(ys)

    if xmax-xmin > ymax-ymin: # y 기준으로 하기
        for y in range(ymin,ymax+1,1):
            some_xs = []
            for p in pts:
                if p[1]==y:
                    some_xs.append(p[0])
            some_xmin = min(some_xs)
            some_xmax = max(some_xs)
            for x in range(some_xmin,some_xmax+1,1):
                canvas[y,x,:] = color

    else: # x 기준으로 하기
        for x in range(xmin,xmax+1,1):
            some_ys = []
            for p in pts:
                if p[0]==x:
                    some_ys.append(p[1])
            some_ymin = min(some_ys)
            some_ymax = max(some_ys)
            for y in range(some_ymin,some_ymax+1,1):
                canvas[y,x,:] = color
    return
```

```python
def main():
    canvas = np.zeros((height,width,3),dtype='uint8')  Canvas 생성
    a,b = width/2,height/2  → Canvas 중앙점
    w,h = 100, 50  → 만들 사각형의 가로, 세로 길이

    while True:
        now = datetime.now()  → 현재시간 갱신.
        hour = now.hour  → 시
        min = now.minute  → 분
        hour_rect = getrectangle(h) → 시침을 위한 사각형
        min_rect = getrectangle(h)  → 분침을 위한 사각형
        center = getRegularNGon(40) → 센터
        circle = getRegularNGon(40) → 큰 시계 틀.

        # print(hour,min)
        center[:,0] *= 15
        center[:,1] *= 15   scale up


        circle[:,0] *= 200
        circle[:,1] *= 200
        # rotating, translating points
        H1 = makeTmat(a,b) @ makeRmat(-90)   중앙에 이동 & 시계로 회전  ▭ → ▯ (12시부터 시작)

        circle = H1 @ circle.T 가운데로 이동
        circle = circle.T
        circle = circle.astype('int')
        drawPolygon(canvas,circle,(255,255,255),axis = False) 큰 시계틀 / (흰색)
        drawColored(canvas,circle,(255,255,255))

        hour_rect = H1 @ makeRmat((hour*30+min/2)) @ makeTmat(0,-h/2) @ hour_rect.T   사각형 가운데 맞추기
        hour_rect = hour_rect.T   30도×hour + 1/2도×min
        hour_rect = hour_rect.astype('int')
        drawPolygon(canvas,hour_rect,(0,255,0),axis = False)
        drawColored(canvas,hour_rect,(0,255,0))  초록색
```

12
6:01
6:00일 30°

360/12 + 다음 시간부터 넘어가려면 분의 360도 앞서야 함 1분 ↓

시침은 1시간에 30도 회전
분침은 1시간에 6도 회전

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